DIGITAL LOGIC

Assignment 2.2

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1). Fill-in-the-blank Questions

1 $F = A \odot B$ **its Minterm expression is** $\bar{A}\,\bar{B}$**+AB**,
its Maxterm expression is $(\boldsymbol{A}+\overline{\boldsymbol{B}})(\bar{A}+\boldsymbol{B})$

2 A function with *n* variables can form 2^n **(how many) Maxterms,** 2^n **(how many) Minterms.**

3 A canonical "AND-OR" expression is made of a number of Minterms by ORing together .

4 F(A,B,C)=∑m(1,2,3,4,7) can be converted to another canonical form, whose abbreviated expression is $\prod M(0,5,6)$

5 If $F(A,B,C) = \prod M(0,2,5)$**, then** $F'(A,B,C) = \sum m(2,5,7)$ **.**

6 The universal gates usually refer to NAND and NOR, such two logic gates.

2). List the "SOM" and "POM" of the following three logic functions（**Do not use the shorthand form**）

1. $\mathrm{F(A,B)} = \sum m(1,3)$**=** $\bar{A}\,\mathrm{B} + AB$

=(A+B)($\bar{A}$ +B)

2. $\mathrm{F(A,B,C)} = \sum m(1,4,6,7) = (\bar{A}\bar{B}C + A\bar{B}\bar{C} + AB\bar{C} + ABC)$

=(A+B+C)(A+$\bar{B} + C$)(A + $\bar{B} + \bar{C}$)($\bar{A} + B + \bar{C}$)

3. $\mathrm{F(A,B,C)} = \prod M(0,1,2,4,5) = (A + B + C)$ **(A**$+B + \bar{C}$**) (A+**$\bar{B} + C$**) (**$\bar{A} + B + C$**) (**$\bar{A} + B + \bar{C}$**)**

= $\bar{A}BC + AB\bar{C}$**+ABC**

- 4 $F(A,B,C,D) = B\overline{C}\overline{D} + \overline{A}B + AB\overline{C}D + BC$

F (A, B, C, D)

$=\sum m(4-7, 12-15)$

$=\prod M(0-3, 8-11)$

□ 5

$$F(A,B,C,D)=\overline{(\overline{A}\,\overline{B}+ABD)}+(B+CD)$$

$$=\overline{\overline{A}\,\overline{B}}\;\overline{ABD}+(B+CD)$$

$$=(A+B)(\overline{A}+\overline{B}+\overline{D})+(B+CD)$$

$$=A+B+CD$$

$$\sum m(3-15)$$

$$=\prod M(0-2)$$